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SCXconvergence

SCX Industry Audit: Multi-Verifier Convergence Analysis Across Six Sectors

SCX

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Abstract

Abstract SCX/YajieManufacturingSemiconductorBankingCivil Servants Corethere exists——Civil Servants$M = 1$BankingManufacturing$M > 1$ISOJORC/NI 43-101 however$M > 1$there exists$\sum g = 0$i.e.allall $M > 1$Conflict of Interestthere exists

Yajie"$M > 1$"Banking/Manufacturing/$M > 1 M > 1 \sum g = 0$——each"" "" $\sum g = 0$SCX"" ""

$\sum g = 0$allSCX5,000/SCX1,000–2,500/

Abstract: This paper presents a systematic analysis of trust-mechanism restructuring under the SCX/Yajie Protocol across six critical industries: professional services, manufacturing, semiconductors, natural resources, banking, and civil service. Our central finding is a structural gradient in trust mechanisms: while services and civil service operate under a near-$M = 1$ self-declaration regime, banking, manufacturing, and resources already have formal $M > 1$ multi-verifier systems (Basel accords, ISO certification, JORC/NI 43-101 qualified-person regimes). However, all existing $M > 1$ systems share a fundamental defect: the absence of the $\sum g = 0$ mathematical constraint—the requirement that the sum of all verifier biases must equal zero and that all verifications must converge toward observable facts. Existing $M > 1$ systems employ multiple verifiers, but verifier biases may systematically lean in the same direction (regulatory capture, paid-auditor conflicts of interest); no mathematical mechanism forces convergence toward truth.

The revolutionary contribution of the Yajie Protocol is not "introducing $M > 1$" (banking/manufacturing/resources already have $M > 1$), but rather **superimposing the $\sum g = 0$ constraint on both existing and new $M > 1$ systems**—transforming each industry's trust mechanism from "multiple verifiers whose biases may coexist" to "multiple verifiers whose biases must sum to zero." $\sum g = 0$ constitutes SCX's genuine technical contribution: it transforms verification from a

sociological question ("who is more credible?") into a mathematical question ("do the biases converge?").

Any industry participant that refuses multi-verifier audit under the $\sum g = 0$ constraint will have all its claims marked as `UNDECLARED`, rendering them non-tradable within the Yajie network. We conservatively estimate the global addressable SCX audit market across six industries at approximately \$500 billion/year, with SCX-capturable share projected at \$100–250 billion/year.

Keywords: SCX protocol, Yajie audit, multi-verifier convergence, $\sum g = 0$ constraint, industry audit, trust mechanism, verifier bias, audit sovereignty, information asymmetry, UNDECLARED marking.

1 / Introduction

1.1 / The Structural Defect of Trust

each however ——Information Asymmetry

Definition 1.1. $\mathcal{A} = \{A_1, A_2, \ldots, A_M\}$ C $M = 1$ $M > 1$

Definition 1.2. $g_i \in \mathbb{R}$ A_i $g_i > 0$ $g_i < 0$ $g_i = 0$

Definition 1.3 ($\sum g = 0$). *Yajieall*

$$\sum_{i=1}^{M} g_i = 0 \tag{1}$$

??

Table 1: / Current Verification Structures Across Six Industries

Industry	**Claimant**	**Claim Content**	M **Current** M	$\sum g = 0$ $\sum g = 0$**?**
/ Services	/ Provider	/ Quality	1	/ No
Manufacturing / Manufacturing	/ / Factory	/ Compliance	≈1	/ No
Semiconductor / Semiconductors	Wafer Fab / Fab	/ Yield	1	/ No
/ Resources	/ Miner	/ Reserves	≈1	/ No
Banking / Banking	/ Bank	/ CAR	>1 ()	/ No
Civil Servants / Civil Service	/ Government	/ Performance	1	/ No

??i.e.BankingManufacturing$M > 1$ISOJORC$\sum g = 0$ ——and——ISO $M > 1\sum g = 0$

1.2 SCX/YajieCore / Core Innovation of SCX/Yajie

Yajie"$M > 1$"——BankingManufacturingthere exists **all**$M > 1\sum g = 0$"" ""

Proposition 1.4 (convergence). $\sum g = 0 A_i g_i \neq 0 M - 1$*because*

$$g_i = -\sum_{j \neq i} g_j \tag{2}$$

i.e.eachallon the contrary$M$

> $\sum g = 0$"" "" —— SCX$\sum g = 0$all`UNDECLARED` convergenceawaitingthere exists each$\sum g = 0$

2 Formal Framework / Formal Framework

2.1 / Multi-Verifier Consensus Model

we formalizeSCX/YajieCore

Definition 2.1. $\mathcal{C} = \{C_1, C_2, \ldots\}$ *alleach* C_k P_k $\theta_k \in \mathbb{R}$ *for example*θ_k

Definition 2.2. A_i C_k

$$\hat{\theta}_{k,i} = \theta_k + g_i + \varepsilon_{k,i} \tag{3}$$

where g_i A_i $\varepsilon_{k,i} \sim \mathcal{N}(0, \sigma_i^2)$

Definition 2.3 (Yajie)**.** $\sum g = 0 M C_k$ *Yajie*

$$\hat{\theta}_k^{Yajie} = \frac{1}{M} \sum_{i=1}^{M} \hat{\theta}_{k,i} \tag{4}$$

$\sum g = 0 \mathbb{E}[\hat{\theta}_k^{Yajie}] = \theta_k + \frac{1}{M} \sum_i g_i = \theta_k$ *i.e. Yajie*

Theorem 2.4 (Bias Detectability)**.** $M \geq 3$ *if* A_i $|g_i| > 0$ *there exists* $1 - \alpha$

$$|g_i| > z_{\alpha/2} \cdot \frac{\sigma_i}{\sqrt{T}} \tag{5}$$

where T $z_{\alpha/2}$

theorem **??** $\sum g = 0$ such that “”——T convergence

2.2 UNDECLARED / Game-Theoretic Analysis of UNDECLARED

Definition 2.5 (UNDECLARED)**.** *if* P_k $M > 1$ *satisfying* $\sum g = 0$ C_k *SCX* `UNDECLAREDUNDECLARED` *SCX*

`UNDECLARED` “i.e.”Silence-as-Admission

Proposition 2.6 (i.e.)**.** *SCXifthere existsYajie* `UNDECLARED` *Bayesian*

$$P(\mid \texttt{UNDECLARED}) > P() \tag{6}$$

there existsif and only if `UNDECLARED` ˛”

`UNDECLARED` “”—— `UNDECLARED`

2.3 / Audit Market Size Estimation

??SCX

Table 2: SCX / Potential SCX Audit Market by Industry

Industry	/ Global Revenue	/ Current Audit %	SCX Potential SCX Market
	~$6.5	~0.5%	~$325
Manufacturing	~$17	~1.0%	~$1,700
Semiconductor	~$6,000	~2.0%	~$120
	~$5	~1.5%	~$750
Banking	~$6	~3.0%	~$1,800
Civil Servants	~$18	~0.3%	~$540
	~$53	—	**~$5,235**

SCX5,000/ i.e.SCXwhere20%1,000/——

3 / Professional Services

3.1 Inflation / Current Structure: Inflation of Declarations

Core“”

- / **Consulting**BCGawaiting“” PPT There is no way to know“Client Renewal Rate”Quality ProofCorporate Inertia——Switching Costs
- / **Legal Services**“”

  ——“”——there existsbecause
- / **Advertising**“” Growth ——
- / **Healthcare** “”
- **Higher Education / Higher Education**“”Proof

  “”——i.e.

  Akerlof——Information Asymmetrybecause

3.2 $\sum g = 0$ / What $\sum g = 0$ Exposes

SCX

(1) / **Decoupling of Brand Premium from Quality Premium** $\sum g = 0$“effectiveness” MProject12
24

SCX

(2) / Case Selection Bias YajieCercisScore ——80%“80%” 50%

(3) / Measurability of Creative Output effectiveness M YajieScoreQuality Proof

(4) / The Truth of Healthcare Quality “” ————YajieScore

(5) “” / Exposure of Educational Value-Added SCX “” YajieScore——

3.3 / Winners and Losers

Table 3: SCX

/ Winners	/ Losers
	Proof
/Information Asymmetry	

3.4 / Recommended Strategy

0–12 SCXYajie“” i.e.SCX$M \geq 536$

12–24 First MoversSCXdefineAudit DimensionsYajieScore“Yajie”

24 YajieFirst Movers——→YajieScore→→

4 Manufacturing / Manufacturing

4.1 / Current Structure: False Security of Single-Point Inspection

Manufacturingassumption**Internal InspectionawaitingQuality Assurance** This assumption is wrong

Manufacturingthere exists——ISO $\sum g = 0$“”

(1) / Incentive Conflict

if Corruption——

(2) / Sampling Bias

(3) / Standard Relaxation define ——“”

for100%for“”

4.2 $\sum g = 0$ / What $\sum g = 0$ Exposes

(1) **reliabilityScale / Magnitude of Self-Inspection Unreliability** simultaneouslyInternal Inspection$MM \geq 3g_i$ 0.1%0.8%——“”Quality Integrity Deficit

(2) **/ Hidden Quality Risk Propagation in Supply Chains** SCXeacheachYajieScore “” “”

(3) **/ Fundamental Flaw of Inspection Economics** —— SCX“”

(4) **ISO / The Ritualistic Nature of ISO Certification** ISO 9001awaiting$M > 1\sum g = 0$ “” SCXISO——$\sum g = 0$

(5) **/ Separation of Process Quality from Inspection Quality**

SCX“” “”——

4.3 / Winners and Losers

Table 4: ManufacturingSCX

/ Winners	**/ Losers**
SPCTQM	“-”
	“”
	—

4.4 / Recommended Strategy

0–12SCX

+Internal Inspection

12–24Yajie allSCX “Yajie”YajieScore

24Yajie YajieCore “Yajie”

5 Semiconductor / Semiconductors

5.1 i.e. / Current Structure: Yield as State Secret

Semiconductor Core——yield——

Core TSMC3nmGAAIntel 4

Wafer Fab“80%”——40%95%awaiting

——Core ifTSMCTSMC Semiconductor“” thus
“” “”
“”

5.2 $\sum g = 0$——Core / What $\sum g = 0$ Exposes—and the Core Tension

(1) $M \sum g = 0$ ——

(2) Semiconductor“” SCX“”

(3) Wafer FabeachAB SCXifWafer Fab

(4) “3.2%”——$M > 1$ “”

Core vs. SemiconductorSCX SCX$M > 1$——SemiconductorCore Wafer Fabawaiting Competitive Advantage

- **AZero-Knowledge AuditZK-Audit**——Zero-Knowledge ProofWafer FabSCXProof ZKSemiconductor
- **BTEE**——Wafer FabTEE
- **C“”**——ifWafer FabSCX`UNDECLARED`
- **D**——Wafer FabSCXWafer Fab

5.3 / Three-Tier Audit Architecture

we propose

Table 5: Semiconductor / Three-Tier Semiconductor Audit Architecture

Level	Description	Application	Valuation
Level 1	Wafer Fab		
Level 2	Abstract		awaiting
Level 3	$M > 1$		

5.4 / Winners and Losers

Winner CategoriesTSMC——“Industry Consensus” “Provable Facts”Zero-Knowledge ProofSemiconductorWafer Fab

Wafer FabInformation AsymmetryMarket Value

5.5 / Recommended Strategy

SemiconductorSCX“” “”

0–24 “Semiconductor”/ZKZK-AuditTEE-Auditfeasibility CoreSCX

24–48 Wafer Fab Level 1Level 2awaitingLevel 3

48 Level 3“” Level 1Wafer Fab“Level 3”

6 / Natural Resources

6.1 / Current Structure: What’s Underground, I Decide

————Core**all**

- **/ Mineral Reserve Declarations** “500LCE”

  there exists**Company ValuationCEO——**

- **/ The Classification Game** “”(Proven)“”(Probable)“”(Possible) JORCNI 43-101 “” “”

- **/ Production Data** —— “”

- **/ Fraudulence of Environmental Claims** “” “” “” ——

- **/ Self-Certification of Carbon Credits** “”

- **/ The Politics of Oil Reserves** OPEC 1980OPEC“”——50%

6.2 $\sum g = 0$ / What $\sum g = 0$ Exposes

(1) / Reserve Bubbles $M > 1$“” $\sum g = 0$if“Inflation”

(2) Decays / The True Grade Decline Curve SCX $\sum g = 0$“”——

(3) / Pre-Disaster Early Warning $M > 1$ ifSCX——“”

(4) “” / The “Authenticity Deficit” of Carbon Credits ProjectSCX++M if60%40%“”Project

(5) / The End of Reserve Politics if1000SCX`UNDECLARED`SCX“” “”

6.3 / Winners and Losers

Table 6: SCX

/ Winners	/ Losers
"Audit Premium"	Inflation
Project	
//	
	—

6.4 / Recommended Strategy

0–18 1–2SCX$M = 5$i.e.——

18–36 SCXalleach/SCXYajie

36Yajie "SCX"SCX-Certified CopperSCX-Certified Lithiumawaiting

7 Banking / Banking

7.1 / Current Structure: Self-Reporting Under Regulatory Rubber-Stamp

Bankingassumption Both parts of this assumption have serious flaws

(1) / Banks' Self-Reporting Incentives

RWA ——1.2%1.0%——"" ——""

(2) / Regulators' Verification Limitations i.e.ECBPRAInformation Asymmetry

(3) / The Performativity of Stress Tests 2008"" assumptionGDP——

(4) auditability / The Unauditability of Complex Financial Instruments ——CDO-CLOETF—— forthere exists""

(5) Banking / The Blind Spot of Systemic Risk ——allsimultaneouslyFails ABi.e.

7.2 $\sum g = 0$ / What $\sum g = 0$ Exposes

(1) $M > 1 \sum g = 0$ all

(2) ""

– ""Capital Integrity Deficit

(3) Non-PerformingNPL "Non-Performing" define "" SCX$M > 1$

(4) LCRNSFR"High-Quality Liquid Assets" define HQLA SCXreliability

(5) ifSCX——

(6) ifSCXBankingthere exists""

7.3 / Winners and Losers

Table 7: BankingSCX

/ Winners	/ Losers
SCX	
SCX	
stability	—

7.4 / Recommended Strategy

0–18 awaitingSCX RWA

18–36 Financial RegulationSCX SCX

36 YajieScore SCX "Yajie" Competitive Advantage——YajieScore

8 Civil Servants / Civil Service

8.1 $M = 1$ / Current Structure: The $M = 1$ Performance Loop

allCivil ServantsInformation Asymmetry——because

(1) $M = 1$→→→"/" Corruption——

(2) **Corruption** Corruption/Corruption CorruptionCorruption Corruptionη——becauseCorruption

(3) **effectiveness** ""

all"" ""——100%

(4)

8.2 $\sum g = 0$ / What $\sum g = 0$ Exposes

(1) Civil Servants$M > 1$———— $\sum g = 0$each ifall$g_{\text{superior}} > 0$——i.e.“”

(2) **Score** AB B$M > 1$ if“YajieScore”

(3) **Corruption** CorruptionSCX—— for example“” there exists$g$–$g > 0$ $\sum g = 0$“CorruptionYajieScor

(4) “”SCX $M > 1$NGO$\sum g = 0$ “Fails”will become provable — which is impossible under the current system

(5) **Scale** SCX——

(6) “XYProject”—— SCX$M > 1$YajieScore

Core Civil ServantsSCXall

- **ASCX**——Civil ServantsSCX/Academic
- **B**——Corruptionawaiting
- **C**——/simultaneouslySCX

8.3 / Winners and Losers

Table 8: Civil ServantsSCX

/ Winners	/ Losers
Civil ServantsProvable CompetencePerceived Loyalty	CorruptionRent-Seeking
	“”
	“”

8.4 / Recommended Strategy

0–24 SCX Corruption

24–48 SCXSCXCivil Servants/

48 SCX“SCX”SCX

9 / Cross-Industry Synthesis

9.1 $M = 1$universality / The Universality of the $M = 1$ Problem

allCore$M = 1$ although

Table 9: $M = 1$ / Unified View of the $M = 1$ Problem Across Six Industries

Industry	Claimant	Claim	Current Verifier
Manufacturing	/		
Semiconductor	Wafer Fab	80%	Wafer Fab
		X	
Banking		15%	
Civil Servants	/	/	

$M = 1$—— SCX/Yajiethere exists$M > 1$ therefore$M = 1$“”——
all“” $M = 1$——“” “”

9.2 $\sum g = 0$ / The Mathematical Violence of $\sum g = 0$

$\sum g = 0$Yajie “”——

“”Failsbecausebecause ————

$\sum g = 0M$ $\sum g = 0$——

——because “” Quality Assurance

9.3 / Structural Sources of First-Mover Advantage

SCXFirst Movers——

(1) / Data Accumulation Positive Feedback First Movers→→YajieScore→→→ First Movers

(2) / Standard-Setting Power First Movers Late MoversFirst MoversdefineAudit Dimensions

(3) / Switching Cost Lock-in YajieScore“” Yajie——

(4) / Talent and Auditor Lock-in / First Movers

9.4 UNDECLARED / The Deterrent Power of UNDECLARED

UNDECLARED SCX “”——

SCXUNDECLARED awaiting“” UNDECLARED “Yajie” “”

UNDECLARED “i.e.” ——

SemiconductorCivil ServantsUNDECLARED //such thatUNDECLARED textbf——“UNDECLARED” “”

9.5 / The New Economic Class of Auditors

SCX**Yajie**

//——$M = 1$ YajieCore

- $g_i \sum g = 0$
- “”——Task
- Consistency
- SCX“”——

9.6 / The Quantifiability of Trust Premium

SCX

SCX/Yajie

- YajieScore10%for
- SCXYajieScore= 0.95forScore= 0.70
- YajieAforB
- YajieScore20%for20%

SCX5–10 “”

9.7 / Cross-Industry Audit Linkages

SCXthere exists

- **-**SCX YajieScoreYajieScore
- **-**YajieScoreYajieScore
- **-Civil Servants** ifCivil ServantsSCXYajieScorethereforeSCX

SCX——

10 / Discussion: Risks and Challenges

10.1 / Auditor Collusion Risk

if$M\sum g = 0$ eachg_i

- M——
- ——
- “”——

10.2 / Privacy-Transparency Balance

SemiconductorCivil Servants ZK-AuditTEE

10.3 / Audit Cost and Coverage

SCX —— for example“Scale”Audit Dimensions$\sum g = 0$ “”

10.4 / Geopolitical Resistance

SCX YajieScore

10.5 “False Positive”Corruption / False Positive Corruption Flagging Risk

Civil ServantsCorruption “Corruption”Civil Servants $p < 0.001$——Corruption

11 Conclusion / Conclusion and Action Framework

11.1 CoreConclusion / Core Conclusions

(1) $\sum g = 0$ Civil Servants$M > 1$BankingManufacturing$M > 1$/ISO/JORC SCX/Yajieall$\sum g = 0$

(2) $\sum g = 0$because“”—— Consistency

(3) **all** —— First Movers

(4) **SemiconductorCivil Servants“”**becauseSemiconductorCivil Servantsthere exists ZK-AuditTEE-Audit

(5) **Yajie** SCX5,000/SCX1,000–2,500/

(6) `UNDECLARED` “i.e.”

(7) **SCXCore** “Provable Integrity”

11.2 / Tiered Action Framework

Table 10: / Tiered Action Framework

Phase	Core Core Actions	Key Milestones
0–12	2–3“”SCX	Yajie
12–36	Top 20	YajieScore
36+	Yajie “”	“” “” SCX

11.3 Concluding Outlook / Concluding Outlook

SCX/Yajie—— It does not change what industries do, but changes how industries**prove they did what they claimed**

SCX

$$\text{“}all \sum g = 0$$

——

becausebecause”

——SCXCore 4.7

“”Trust-Me SCX/Yajie“”Verify-Me —— Configurationprecision ——becausebecause

A AppendixATerm / Appendix A: Glossary

Term / Term	**define / Definition**
M	$/M = 1M > 1$
g_i	$ig_i > 0g_i < 0$
$\sum g = 0$	YajieCoreall

Term / Term	**define / Definition**
YajieScore / Yajie Score	$\sum g = 0M$
Cercis	YajieScoreScore
`UNDECLARED`	SCX$M > 1$
/ Trust Premium	/
ZK-Audit	Zero-Knowledge AuditProofsatisfying
TEE-Audit	
/ Capital Integrity Deficit	
/ Quality Integrity Deficit	
Inflation / Reserve Inflation Factor	
i.e. / Silence-as-Admission	there exists`UNDECLARED`

Table 11: Term / Glossary of Terms

B AppendixBAudit Dimensions / Appendix B: Industry Audit Dimensions Reference

B.1 Audit Dimensions / Services Audit Dimensions

	CoreAudit Dimensions	M	
	-	5+	Project12–24
		5+	
		5+	Project
		7+	
Higher Education	-	5+	

B.2 ManufacturingAudit Dimensions / Manufacturing Audit Dimensions

	CoreAudit Dimensions	M
	vs	3+
stability	Cpk	3+
		3+
	YajieScore	N/A

B.3 SemiconductorAudit Dimensions / Semiconductor Audit Dimensions

	CoreAudit Dimensions		
$<$5nm		Level 12	TEE-Audit
$\geq$7nm		Level 23	+
	stability	Level 3	

B.4 Audit Dimensions / Resources Audit Dimensions

	CoreAudit Dimensions	M	
	/	5+	+
		5+	+
		5+	/
	vs	5+	

B.5 BankingAudit Dimensions / Banking Audit Dimensions

	CoreAudit Dimensions	M
	RWANPL	5+
	VaR	5+
	LCRNSFR	5+
		3+

B.6 Civil ServantsAudit Dimensions / Civil Service Audit Dimensions

	CoreAudit Dimensions	M	
		5+	Project
		5+	
		5+	
		5+	/
		7+	
Corruption	-	7+	

C AppendixC / Appendix C: Key Mathematical Derivations

C.1 Bias DetectabilityProof / Proof of Bias Detectability (Theorem ??)

Proof. $A_i T$

$$\hat{\theta}_{k,i} = \theta_k + g_i + \varepsilon_{k,i}, \quad \varepsilon_{k,i} \sim \mathcal{N}(0, \sigma_i^2) \tag{7}$$

$\sum g = 0 M - 1$

$$\bar{\theta}_{k,-i} = \frac{1}{M-1} \sum_{j \neq i} \hat{\theta}_{k,j} = \theta_k + \frac{1}{M-1} \sum_{j \neq i} g_j + \bar{\varepsilon}_{k,-i} \tag{8}$$

where$\bar{\varepsilon}_{k,-i} \sim \mathcal{N}(0, \sigma_{-i}^2/(M-1))\sigma_{-i}^2 = \frac{1}{M-1} \sum_{j \neq i} \sigma_j^2$
define

$$D_{k,i} = \hat{\theta}_{k,i} - \bar{\theta}_{k,-i} = g_i - \frac{1}{M-1} \sum_{j \neq i} g_j + (\varepsilon_{k,i} - \bar{\varepsilon}_{k,-i}) \tag{9}$$

$\sum g = 0 g_i = -\sum_{j \neq i} g_j$therefore

$$\frac{1}{M-1} \sum_{j \neq i} g_j = -\frac{g_i}{M-1} \tag{10}$$

$$D_{k,i} = g_i + \frac{g_i}{M-1} + \eta_{k,i} = \frac{M}{M-1} g_i + \eta_{k,i} \tag{11}$$

where$\eta_{k,i} \sim \mathcal{N}(0, \sigma_i^2 + \sigma_{-i}^2/(M-1))$
T

$$\bar{D}_i = \frac{1}{T} \sum_{k=1}^{T} D_{k,i} = \frac{M}{M-1} g_i + \bar{\eta}_i, \quad \bar{\eta}_i \sim \mathcal{N}\left(0, \frac{\sigma_i^2 + \sigma_{-i}^2/(M-1)}{T}\right) \tag{12}$$

$H_0 : g_i = 0$

$$Z = \frac{\bar{D}_i}{\sqrt{(\sigma_i^2 + \sigma_{-i}^2/(M-1))/T}} \sim \mathcal{N}(0,1) \tag{13}$$

H_0 $|Z| > z_{\alpha/2}$i.e.

$$\left|\frac{M}{M-1} g_i\right| > z_{\alpha/2} \cdot \sqrt{\frac{\sigma_i^2 + \sigma_{-i}^2/(M-1)}{T}} \tag{14}$$

$T \to \infty \to 0$thereforeanyg_i □

C.2 i.e. / Formal Game Model of Silence-as-Admission

Proof. $P\tau \in \{H, L\}$where$H$“” L“” $PP(\tau = H) = pP(\tau = L) = 1 - p$

PSCXar if$\tau = H$1if$\tau = L$1

HR_H0 $L{-}C0$

$HaLr$ r

$$P(\tau = L \mid r) = \frac{P(r \mid L) \cdot P(L)}{P(r \mid L) \cdot P(L) + P(r \mid H) \cdot P(H)} = \frac{1 \cdot (1-p)}{1 \cdot (1-p) + 0 \cdot p} = 1 \tag{15}$$

therefore$P(\mid \texttt{UNDECLARED}) = 1 > P() = 1 - p$forany$p < 1$ □

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